Cache Design

CS 2410

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Oracle Cache

- CPU Cycles = CPU Compute Cycles + Memory Stall Cycles

- **Oracle cache**: a cache that never misses
  - In effect, **Memory Stall Cycles == 0**
  - Impossible, since even with infinite capacity, there are still cold misses
  - But useful to set **bounds** on performance

- Real caches may approach performance of oracle caches but can't exceed

- What metric can we use to compare and evaluate real cache designs?
  - AMAT (Average Memory Access Time)

AMAT (Average Memory Access Time)

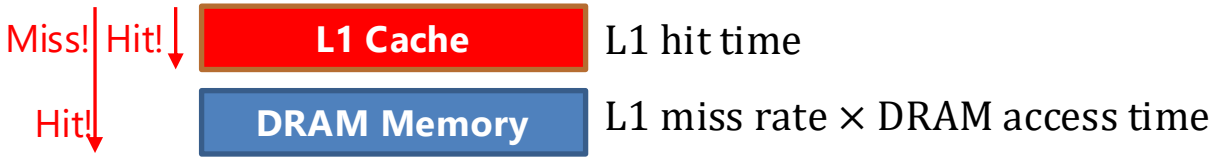
- **AMAT** (Average Memory Access Time) is defined as follows:
  - **AMAT = hit time + (miss rate × miss penalty)**
  - **Hit time:** time to get the data from cache when we hit
  - **Miss rate:** what percentage of cache accesses we miss
  - **Miss penalty:** time to get the data from lower memory when we miss
  - Shouldn't it be **hit rate × hit time** instead of just **hit time**?
    - Hit time is incurred regardless of hit or miss
    - It is more aptly called access time (the time to search for the data)

- Hit time, miss rate, miss penalty are the 3 components of a cache design
  - When evaluating a cache design, we need to consider all 3
  - E.g. A larger cache trade-offs longer hit time for smaller miss rate

AMAT of No-Cache vs. L1 Cache

- For a CPU with no caches:
  - $\text{AMAT(none)} = \text{DRAM access time}$

- For a single-level cache (L1 cache):
  - $\text{AMAT(L1)} = \text{L1 hit time} + (\text{L1 miss rate} \times \text{DRAM access time})$



AMAT of No-Cache vs. L1 Cache

- For L1 Cache to be worth it, AMAT(None) > AMAT(L1) needs to be true.

DRAM Memory	DRAM access time
>?	
L1 Cache	L1 hit time
DRAM Memory	L1 miss rate × DRAM access time

- $\text{AMAT(None)} > \text{AMAT(L1)}?$

$\rightarrow$ DRAM access time > L1 hit time + L1 miss rate × DRAM access time

$\rightarrow$ (1 – L1 miss rate) × DRAM access time > L1 hit time

$\rightarrow$ Benefit from reduced DRAM accesses > Penalty from accessing L1

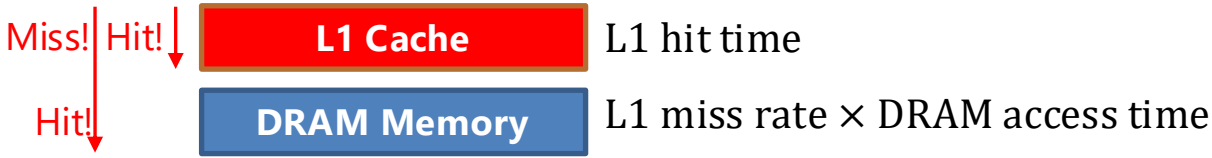
- So, should we add an L1 cache or not?
  - Depends on L1 miss rate, which depends on locality present in software!

AMAT of No-Cache vs. L1 Cache

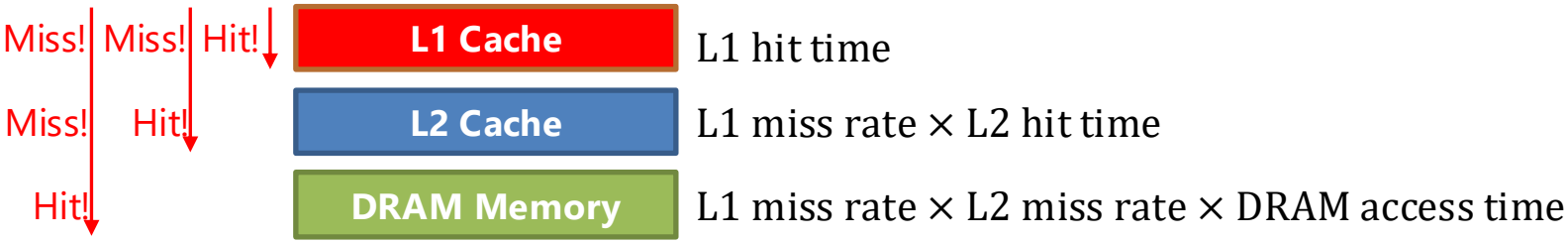
- Let's say CACTI simulation shows a 32 KB L1 cache is feasible with 1 cycle delay
- For L1 cache to be worth it: $(1 - \text{L1 miss rate}) \times \text{DRAM access time} > \text{L1 hit time}$
- If DRAM access time = 100 cycles:
  $(1 - \text{L1 miss rate}) \times 100 > 1 \Rightarrow \text{L1 miss rate} < 0.99$
  - If L1 miss rate can be kept below 99% with 32 KB cache, worth it!
- If DRAM access time = 10 cycles:
  $(1 - \text{L1 miss rate}) \times 10 > 1 \Rightarrow \text{L1 miss rate} < 0.9$
  - If L1 miss rate can be kept below 90% with 32 KB cache, worth it!
- If DRAM access time = 1 cycle:
  $(1 - \text{L1 miss rate}) \times 1 > 1 \Rightarrow \text{L1 miss rate} < 0$
  - The L1 cache will only cause additional delays due to hit time

AMAT of Only L1 Cache vs. L1 + L2 Cache

- For a single-level cache (L1 cache):
  - $\text{AMAT} = \text{L1 hit time} + (\text{L1 miss rate} \times \text{DRAM access time})$



- For a multi-level cache (L1, L2 caches):
  - $\text{AMAT} = \text{L1 hit time} + (\text{L1 miss rate} \times \text{AMAT(L2)})$
  - $\text{AMAT(L2)} = \text{L2 hit time} + (\text{L2 miss rate} \times \text{DRAM access time})$
  - $\text{AMAT} = \text{L1 hit time} + \text{L1 miss rate} \times \text{L2 hit time}$
    $+ \text{L1 miss rate} \times \text{L2 miss rate} \times \text{DRAM access time}$



Cache Design Parameter 1: Cache Capacity

Impact of Cache Capacity on AMAT

- AMAT = hit time + (miss rate × miss penalty)

- Larger caches are **good** for **miss rates**
  - More capacity means you can keep around cache blocks for longer
  - Means you can leverage more of the pre-existing **temporal locality**
  - If entire working set can fit into the cache, no capacity misses!

- But larger caches are **bad** for **hit times**
  - Longer wires and larger decoders mean longer access time

Impact of resizing the L1 cache

- Let's first consider L1 capacity.

L1 Cache	L1 hit time
L2 Cache	L1 miss rate × L2 hit time
DRAM Memory	L1 miss rate × L2 miss rate × DRAM access time

>?

L1′ Cache	L1′ hit time
L2 Cache	L1′ miss rate × L2 hit time
DRAM Memory	L1′ miss rate × L2 miss rate × DRAM access time

- $\mathrm{AMAT}(\mathrm{L1}_{\mathrm{old}}) - \mathrm{AMAT}(\mathrm{L1}_{\mathrm{new}})$
  $= \mathrm{L1\ hit\ time} - \mathrm{L1'\ hit\ time} + (\mathrm{L1\ miss\ rate} - \mathrm{L1'\ miss\ rate}) \times \mathrm{AMAT}(\mathrm{L2})$
  (With the simplifying assumption that L2 miss rate stays the same)

Impact of resizing the last level cache

- Let's first consider L2 capacity.

L1 Cache	L1 hit time
L2 Cache	L1 miss rate × L2 hit time
DRAM Memory	L1 miss rate × L2 miss rate × DRAM access time

>?

L1 Cache	L1 hit time
L2′ Cache	L1 miss rate × L2′ hit time
DRAM Memory	L1 miss rate × L2′ miss rate × DRAM access time

- $\mathrm{AMAT}(\mathrm{L2}_{\mathrm{old}}) - \mathrm{AMAT}(\mathrm{L2}_{\mathrm{new}})$
  $= \mathrm{L2\ hit\ time} - \mathrm{L2'\ hit\ time} + (\mathrm{L2\ miss\ rate} - \mathrm{L2'\ miss\ rate}) \times \mathrm{AMAT(DRAM)}$

Comparing resizing L1 and last level cache

- To recap…
- $\mathrm{AMAT}(\mathrm{L1}_{\mathrm{old}}) - \mathrm{AMAT}(\mathrm{L1}_{\mathrm{new}}) > 0$ equates to:
  $\mathrm{L1\ hit\ time} - \mathrm{L1'\ hit\ time} + (\mathrm{L1\ miss\ rate} - \mathrm{L1'\ miss\ rate}) \times \mathrm{AMAT}(\mathrm{L2}) > 0$
- $\mathrm{AMAT}(\mathrm{L2}_{\mathrm{old}}) - \mathrm{AMAT}(\mathrm{L2}_{\mathrm{new}}) > 0$ equates to:
  $\mathrm{L2\ hit\ time} - \mathrm{L2'\ hit\ time} + (\mathrm{L2\ miss\ rate} - \mathrm{L2'\ miss\ rate}) \times \mathrm{AMAT}(\mathrm{DRAM}) > 0$
- Since AMAT(L2) < AMAT(DRAM)
  - Hit time is more important for L1 → smaller cache
  - Miss rate is more important for L2 → larger cache

What cache size(s) should I choose?

- How should each cache level be sized?

- **Ideally**, cache levels and sizes would **reflect working set sizes**.
  - So that each level of cache leverage a level of temporal locality. E.g.:
    - L1: Small set of data accessed very frequently (typically stack variables)
    - L2: Medium set of data (actively accessed part of data structure)
    - L3: Larger set of data (entire data structure)
    - Memory: All program data

- Simulate multiple cache levels and sizes and choose one with lowest AMAT
  - Simulate on the applications that you care about
  - In the end, it must be a **compromise** (giving best average AMAT)

Cache Design Parameter 2: Cache Block Size

Impact of Cache Block Size on AMAT

- AMAT = hit time + (miss rate × miss penalty)

- **Cache block** (a.k.a. **cache line**)
  - Unit of transfer for cache data (typically 32 or 64 bytes)
  - If program accesses any byte in cache block, entire block is brought in
  - Each level of a multi-level cache can have a different cache block size

- Impact of larger cache block size on **miss rate**
  - Maybe **smaller miss rate** due to **better** leveraging of **spatial locality**
  - Maybe **bigger miss rate** due to **worse** leveraging of **temporal locality** (Bringing in more data at a time may push out other useful data)

- Impact of larger cache block size on **miss penalty**
  - With a limited bus width, may take multiple transfers for a large block
  - E.g. DDR 4 DRAM bus width is 8 bytes, so 8 transfers for 64-byte block
  - Could lead to **increase in miss penalty**

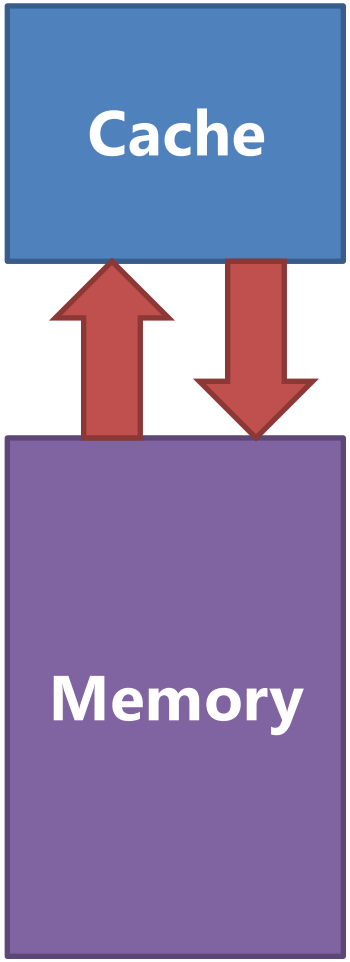
Cache Block Size impact on Miss Penalty

- On a miss, the data must come from lower memory
- Block size determines block transfer time:

$$\frac{\text{seconds}}{\text{block}} = \underbrace{\frac{\text{seconds}}{\text{byte}}}_{\text{byte transfer time}} \times \underbrace{\frac{\text{bytes}}{\text{block}}}_{\text{block size}}$$

$$= \frac{1}{\underbrace{\frac{\text{cycles}}{\text{second}}}_{\text{bus cycle time}} \times \underbrace{\frac{\text{bytes}}{\text{cycle}}}_{\text{bus width}}} \times \underbrace{\frac{\text{bytes}}{\text{block}}}_{\text{block size}}$$

- Has both latency and bandwidth implications



What cache block size should I choose?

- Again, that depends on the application
  - How much spatial and temporal locality the application has

- Simulate multiple cache block sizes and choose one with lowest AMAT
  - Simulate on benchmarks that you care about and choose best average
  - You may end up with different block sizes for each level of the cache

Cache Design Parameter 3: Cache Associativity

Mapping blocks from memory to caches

- Cache size is much smaller compared to the entire memory space
  - Must map all the blocks in memory to limited CPU cache
- Does this sound familiar? Remember branch prediction?
  - Had similar problem of mapping PCs to a limited BHT
  - What did we do then?
    - We hashed PC to an entry in the BHT
    - On a hash conflict, we replaced old entry with more recent one

- We will use a similar idea with caches
  - **Hash memory addresses** to entries in cache
  - On a conflict:
    - **Replace** old cache block with more recent one
    - Or, **chain** multiple cache blocks on to same hash entry

Impact of Cache Associativity on AMAT

- Depending on hash function and chaining, a cache is either:
  - **Direct-mapped** (**no chaining** allowed)
  - **Set-associative** (**some chaining** allowed)
  - **Fully-associative** (**limitless chaining** allowed)

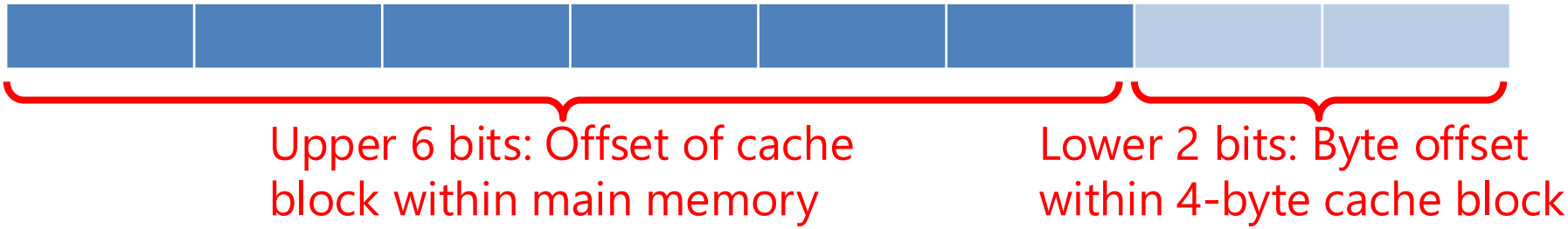
- Impact of more associativity on **miss rate**
  - **Smaller miss rate** due to less misses due to hash conflicts
  - Misses due to hash conflicts are called **conflict misses**
    - A third category of misses besides cold and capacity misses

- Impact of more associativity on **hit time**
  - **Longer hit time** due to need to search through long chain

Direct-mapped Caches

Assumptions

- Let's assume for the sake of concise explanations
  - 8-bit memory addresses
  - 4-byte (one word) cache block sizes
- Of course these are not typical values. Typical values are:
  - 32-bit or 64-bit memory addresses (32-bit or 64-bit CPU)
  - 32-byte or 64-byte cache blocks sizes (for spatial locality)
  - But too many bits in addresses are going to give you a headache
- According to our assumption, here's a breakdown of address bits



  - When I refer to addresses, I will sometimes omit the lower 2 bits (When we talk about cache block transfer, that part is irrelevant)

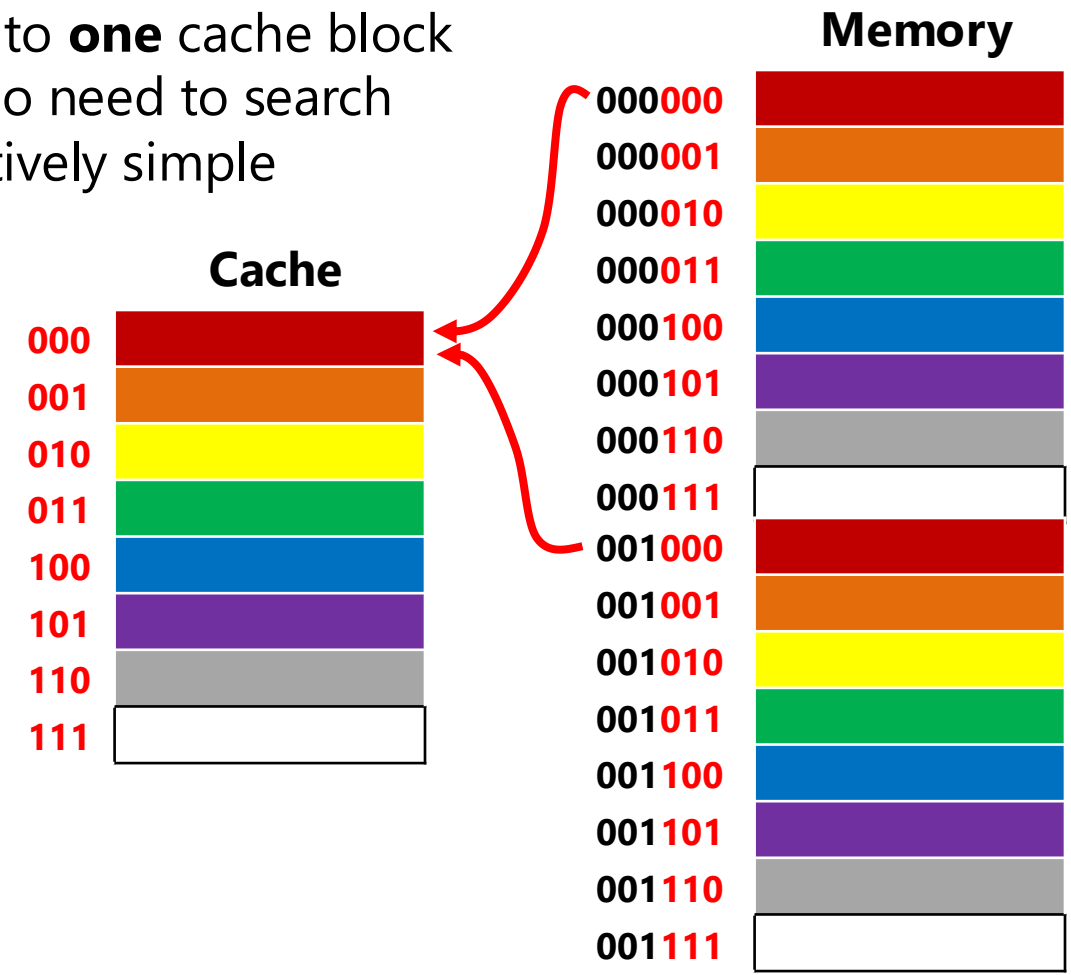
Direct-mapped Cache Hash Function

- Each memory address maps to **one** cache block
  - No chaining allowed so no need to search
  - Implementing this is relatively simple

Hash function:
For this 8-entry cache, to find **cache block index**, take the lowest 3 cache block offset bits in address.

But if our program accesses **001000**, then **000000**, how do we tell them apart?

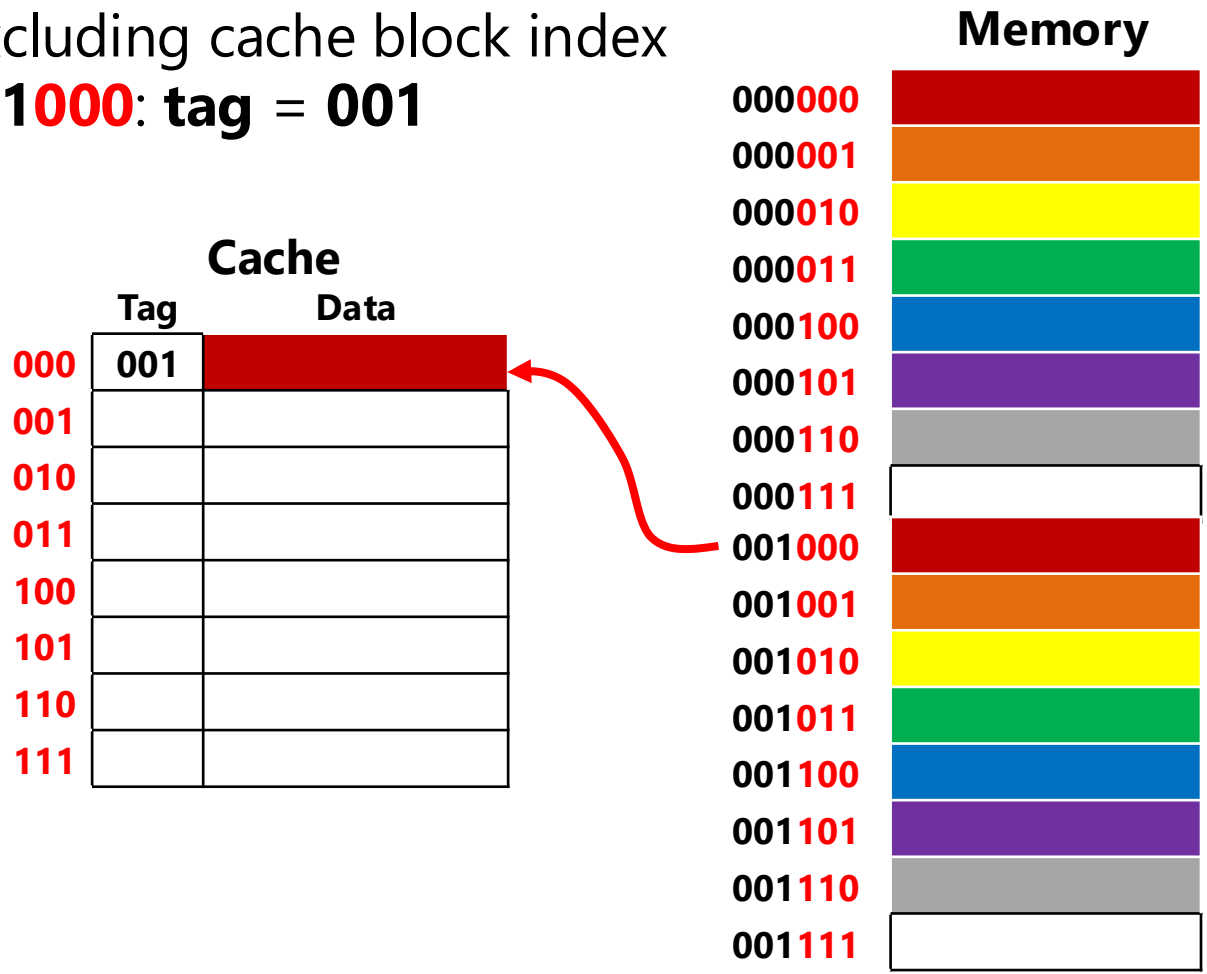
Tags!



Tags help differentiate between conflicting blocks

Tag: part of address excluding cache block index

- On allocation of **001000**: **tag** = **001**



Tags help differentiate between conflicting blocks

Tag: part of address excluding cache block index

- On allocation of **001000**: **tag** = **001**
- On allocation of **000000**: **tag** = **000**

Cache

	Tag	Data
000	000	
001		
010		
011		
100		
101		
110		
111		

Memory

000000	
000001	
000010	
000011	
000100	
000101	
000110	
000111	
001000	
001001	
001010	
001011	
001100	
001101	
001110	
001111	

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Valid bit indicates that block contains valid data

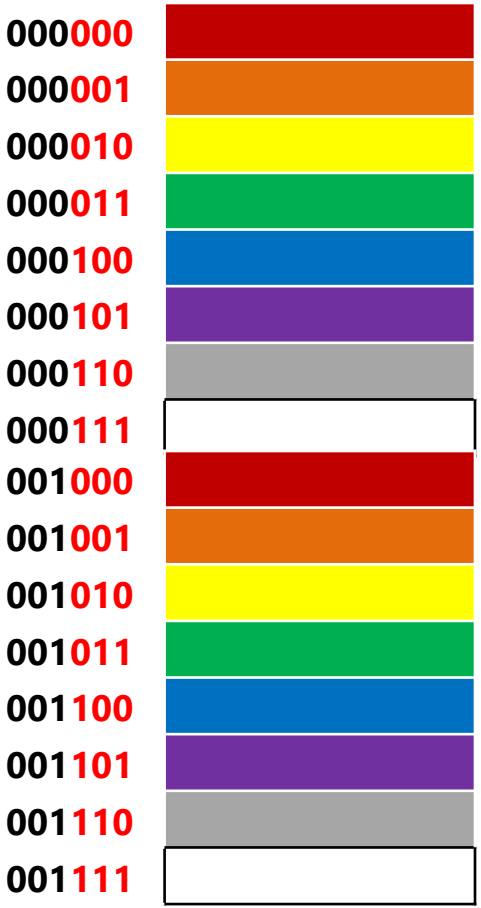
Valid bit: indicates that the block is valid

- Set to 0 initially when cache block is empty
- Set to 1 when a cache block is allocated

Cache

	V	Tag	Data
000	1	000	
001	0		
010	0		
011	0		
100	0		
101	0		
110	0		
111	0		

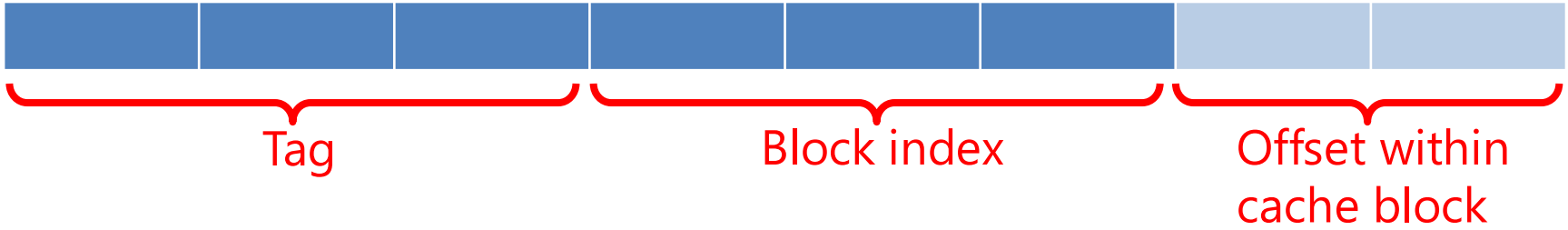
Memory



- Cache hit: V == 1 &&
  CacheBlock.Tag == MemoryBlock.Tag

Quiz: Address Bits Breakdown

- Now with the following parameters:
  - 8-bit memory addresses
  - 4-byte cache block sizes
  - 8-block cache
- How would we breakdown the memory address bits?



  - First, the correct cache block is accessed using the **block index**
  - Then, the **tag** is compared to the cache block tag
  - If matched, **offset** is used to access specific byte within block

Example: A Direct-mapped Cache

- When the program first starts, we **set all the valid bits to 0**.
  - Signals all cache lines are empty
- Now let's try a sequence of reads... do these **hit** or **miss?** How do the cache contents change?

000000	**miss**	Cold misses
100101	**miss**	
100110	**miss**	
100101	**hit**	
010000	**miss**	← Cold miss
000000	**miss**	← Capacity miss?

tag index

	V	Tag	Data
000	**1**	**010**	**something**
001	**0**		
010	**0**		
011	**0**		
100	**0**		
101	**1**	**100**	**something**
110	**1**	**100**	**something**
111	**0**		

Conflict Misses

- What should we call 2nd miss on **000000**?
  - Awkward to call it a capacity miss (It's not like capacity was lacking)
  - Let's call it a **conflict miss**

000000 **miss**
100101 **miss**
100110 **miss**
} Cold misses

100101 **hit**

010000 **miss** ← Cold miss

000000 **miss** ← Capacity miss? Conflict miss!

tag **index**

	V	Tag	Data
000	1	010	something
001	0		
010	0		
011	0		
100	1	100	something
101	1	100	something
110	0		
111	0		

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Types of Cache Misses (Revised)

- Besides cold misses and capacity misses, there are conflict misses

- **Cold miss** (a.k.a. **compulsory miss**)
  - Miss suffered when data is accessed for the **first time** by program

- **Capacity miss**
  - Miss on a **repeat access** suffered due to a lack of **capacity**
  - When the program's **working set is larger than can fit in the cache**

- **Conflict miss**
  - Miss on a **repeat access** suffered due to a lack of **associativity**
  - **Associativity**: degree of freedom in associating cache block with address
  - Direct mapped caches have no associativity
    - Since cache blocks are directly mapped to a particular block index

Associative caches

Flexible block placement

- Direct-mapped caches can have lots of **conflicts**
  - Multiple memory locations "fight" for the same cache line
- Assume:
  1. Cache is 4-block direct-mapped
  2. Following strided memory access pattern:
     - 0, 16, 32, 48
     - 00000000, 00010000, 00100000, 00110000

	V	Tag	Data
00	1	0011	
01	0		
10	0		
11	0		

- **What would happen?**
  - They all land on the same index, and kick each other out.
  - Those other 3 blocks are not even getting used!
  - What if we used the space to chain conflicting blocks?

Full associativity

- Let's make our 4-block cache **4-way set-associative**.

V	Tag	D
1	000000	

V	Tag	D
1	001100	

V	Tag	D
1	000100	

V	Tag	D
1	001000	

- What's the difference?
  - Now a hashed location can be associated with **any** of the 4 blocks
  - Analogous to having a hash conflict chain 4-entries long
  - The 4 cache blocks are said to be part of a cache **set**
  - When set size == cache size, it is said to be **fully associative**
- Let's do that sequence of reads again: 0, 16, 32, 48, 0, 16, 32, 48...
- Notice tag is now bigger, since no index bits are needed for selection
  - Index is called **set index** in this context (since it designates a set)
- Now cache holds the entire **working set**: no more misses!

Example: A 2-way Set-Associative Cache

- 8-block 2-way set-associative cache (same size but more associative)
- Let's try the same stream of accesses as direct-mapped cache
- Yay! 2nd access to **000000** is no longer a conflict miss!

000000 miss
100101 miss
100110 miss
100101 hit
010000 miss
000000 hit

tag index

Set	V	Tag	Data	V	Tag	Data
00	1	0000	something	1	0100	something
01	1	1001	something	0		
10	1	1001	something	0		
11	0			0		

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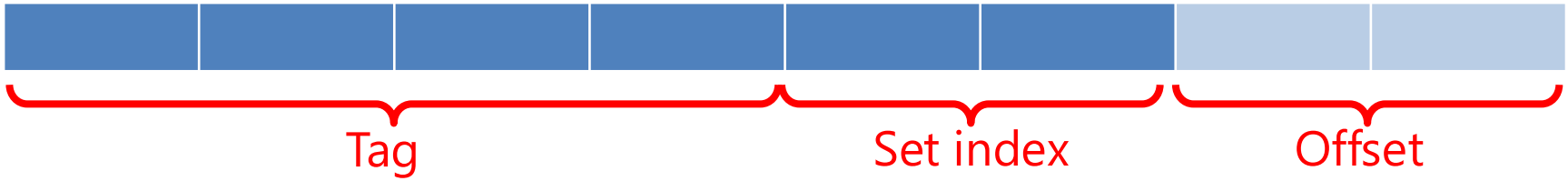
Address Bits Breakdown

- A fully associative cache (doesn't matter how many blocks):

- With 16-block 4-way set-associative cache:



  - 16 / 4 = 4 sets in cache. So, 2 bits required for set index.
- With 64-block 4-way set-associative cache:

  - 64 / 4 = 16 sets in cache. So, 4 bits required for set index.

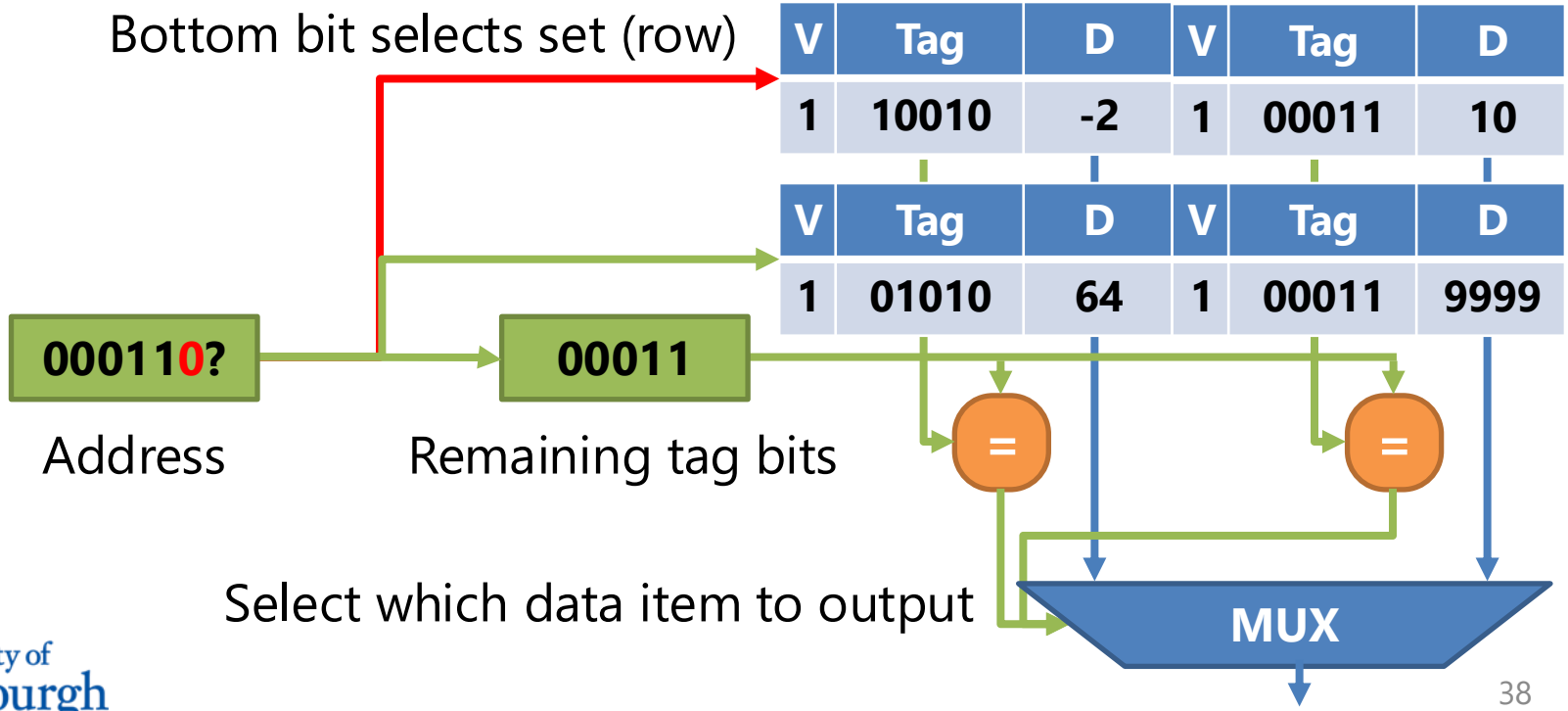
Want More Examples?

- Try out the Cache Visualizer on the course github:
  - https://github.com/wonsunahn/CS2410_Spring2026/tree/main/resources/cache_demo
  - Courtesy of Jarrett Billingsley

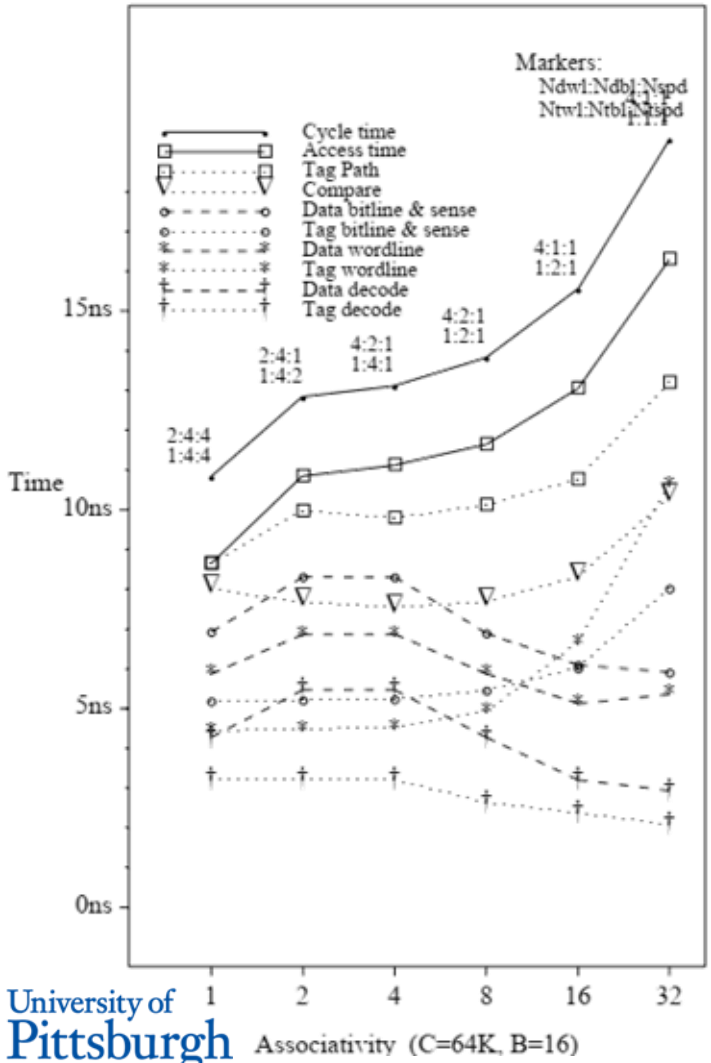
- Visualizes cache organization for various parameters
  - Cache block size
  - Number of blocks in cache (capacity)
  - Cache associativity

Associativity is Costly

- Associativity requires complex circuitry and **increases hit time**
- Full associativity only used when a miss is extremely costly
- Typically, caches are direct mapped or 2-, 4-, 8-way set-associative



Access/cycle time as a function of associativity



Steven J.E. Wilton and Norman P. Jouppi, "An Enhanced Access and Cycle Time Model for On-Chip Caches", WRL Research Report 93/5, 1994.

- First paper to introduce **CACTI**, a simulator for memory structures
  - Still used today to model caches
- Components that increase delay
  - Compare (of tags)
  - Tag wordline & bitline delays

Cache Design Parameter 4: Cache Replacement Policy

Cache Replacement

- If we have a cache miss and no empty blocks, what then?

V	Tag	D
1	000000	

V	Tag	D
1	001100	

V	Tag	D
1	000001	

V	Tag	D
1	001000	

- Let's read memory address 4 (**000001**00).
  - Uh oh. That's a miss. Where do we put it?
- With associative caches, you must have a **replacement scheme.**
  - Which block to evict (kick out) when you're out of empty slots?
- The simplest replacement scheme is **random**.
  - Just pick one. Doesn't matter which.
- What would make more sense?
  - How about taking **temporal locality** into account?

LRU (Least-Recently-Used) Replacement

- When you need to evict a block, kick out the oldest one.

V	Tag	Age
1	000000	3

V	Tag	Age
1	001100	0

V	Tag	Age
1	000100	2

V	Tag	Age
1	001000	1

- Our read history is: 000000, 000100, 001000, 001100.
- Our next read is to 000001. Which block should we replace?

LRU (Least-Recently-Used) Replacement

- When you need to evict a block, kick out the oldest one.

V	Tag	Age
1	000001	0

V	Tag	Age
1	001100	1

V	Tag	Age
1	000100	3

V	Tag	Age
1	001000	2

- Our read history is: 000000, 000100, 001000, 001100.
- Our next read is to 000001. Which block should we replace?
  - **Least recently used** since it is not likely to be used any time soon. (Assuming that the program has **temporal locality**.)
  - On a cache hit, reset counter to 0 (most recently used)
  - For other blocks in set, age by incrementing saturating counter

Impact of LRU on AMAT

- AMAT = hit time + (miss rate × miss penalty)

- Impact of LRU on **miss rate**
  - **Smaller miss rate** due to better leveraging of **temporal locality**
    Larger miss rate due to decrease in capacity due to **added metadata**
    - Saturating counter for LRU uses bits and adds to metadata
    - The **valid bit**, **tag**, and **saturating counter** are all metadata
    - Additional bits for metadata comes in expense of bits for real data
      - Need to limit the number of bits in counter
      - But with reduced bits counter will saturate early, reducing accuracy

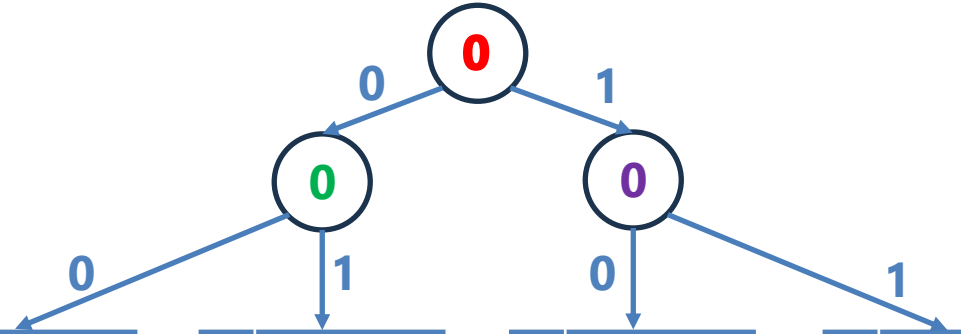
- Modern CPUs use schemes like PLRU (Pseudo-LRU) to save on bits

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Each bit in PLRU field correspond to nodes in binary tree
  - Tells direction to traverse to find the Least Recently Used block



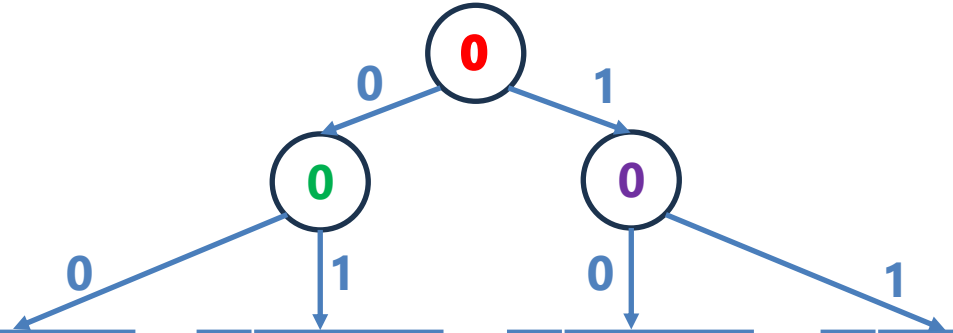
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	1	000000	1	000001	1	000010	1	000011

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000000. Which block should we choose?



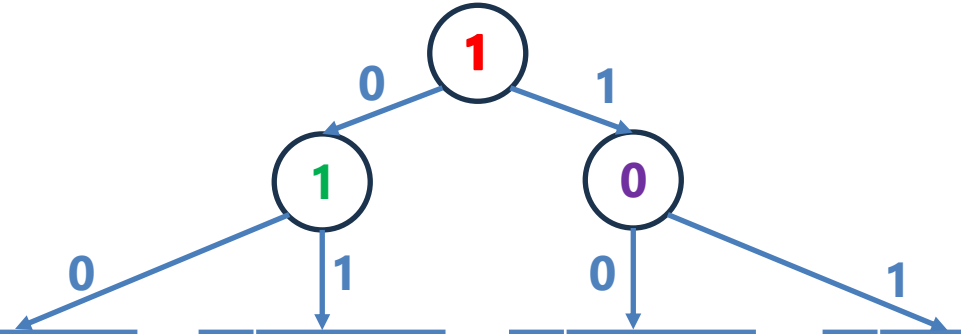
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	1	000000	0		0		0	

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000000. Which block should we choose?
  - Then update PLRU bits so they point opposite direction traversed.



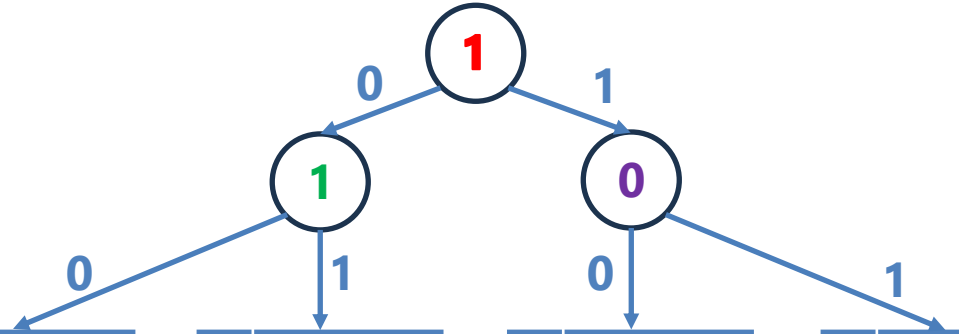
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
110	1	000000	0		0		0	

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000001. Which block should we choose?



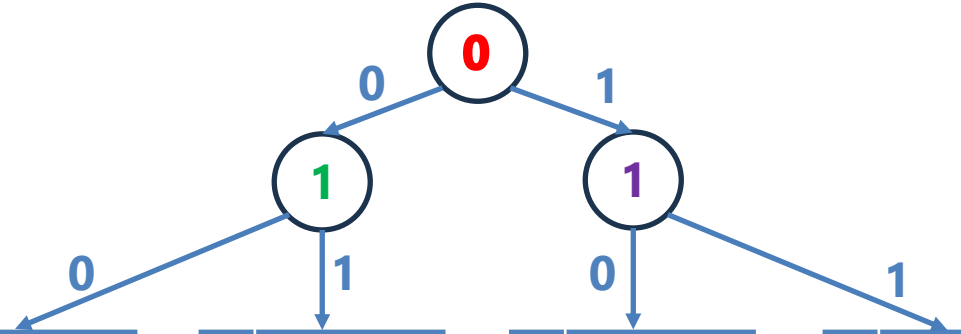
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
110	1	000000	0		1	000001	0	

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000001. Which block should we choose?
  - Then update PLRU bits so they point opposite direction traversed.



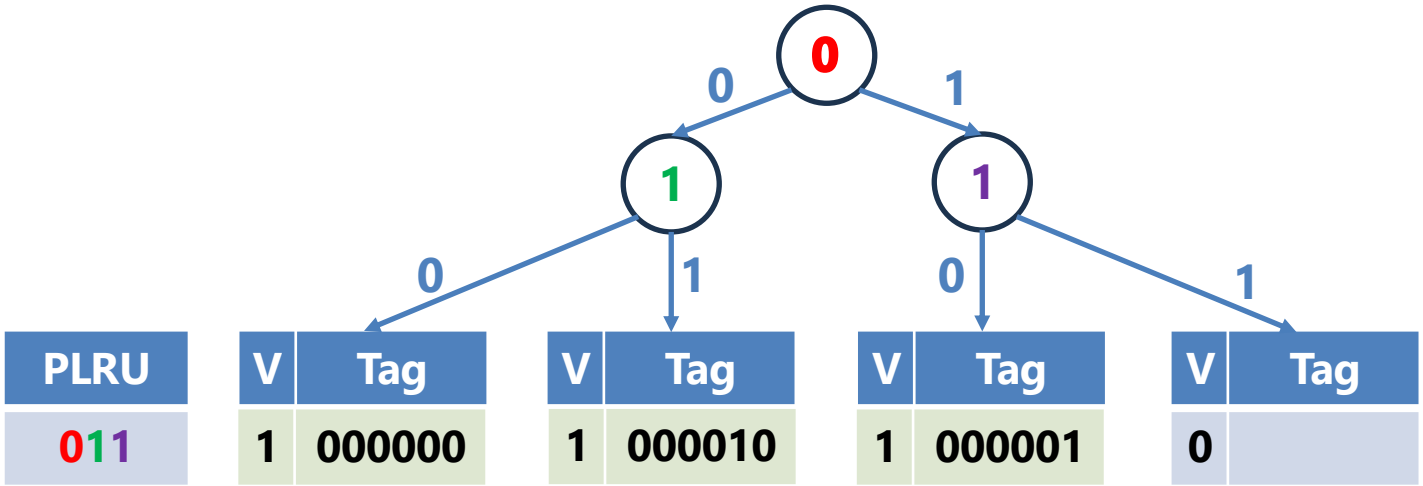
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
011	1	000000	0		1	000001	0	

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000010. Which block should we choose?

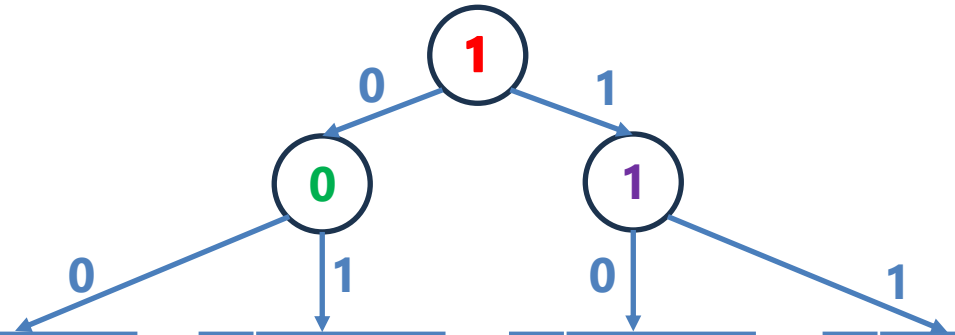


PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000010. Which block should we choose?
  - Then update PLRU bits so they point opposite direction traversed.



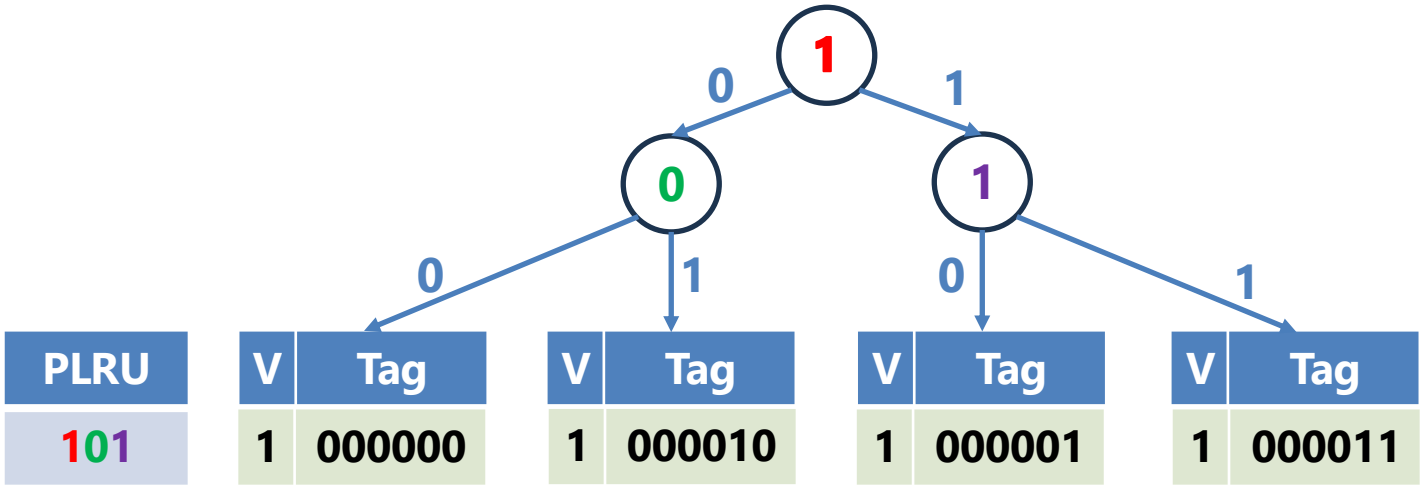
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
101	1	000000	1	000010	1	000001	0	

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000011. Which block should we choose?

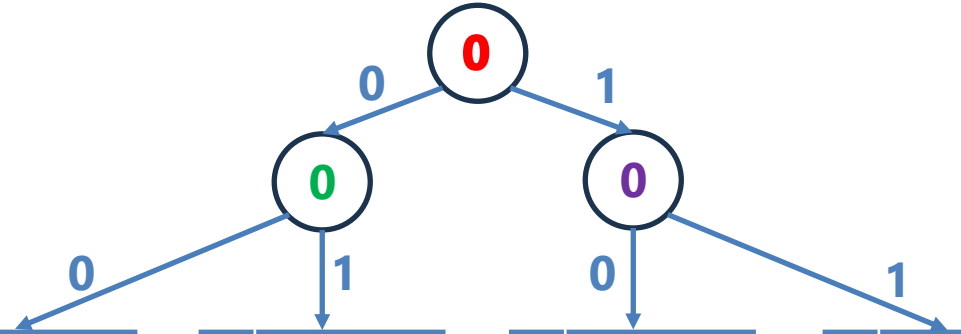


PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Our next read is to 000011. Which block should we choose?
  - Then update PLRU bits so they point opposite direction traversed.



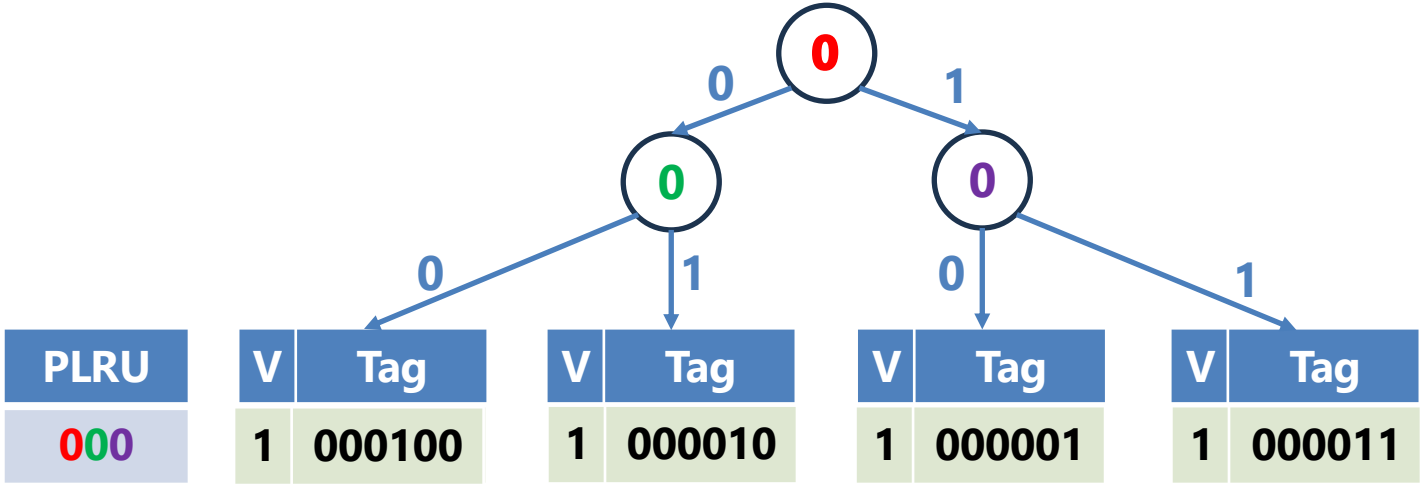
PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	1	000000	1	000010	1	000001	1	000011

PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Say our next read is to 000100.
  - Then we correctly choose the LRU block!

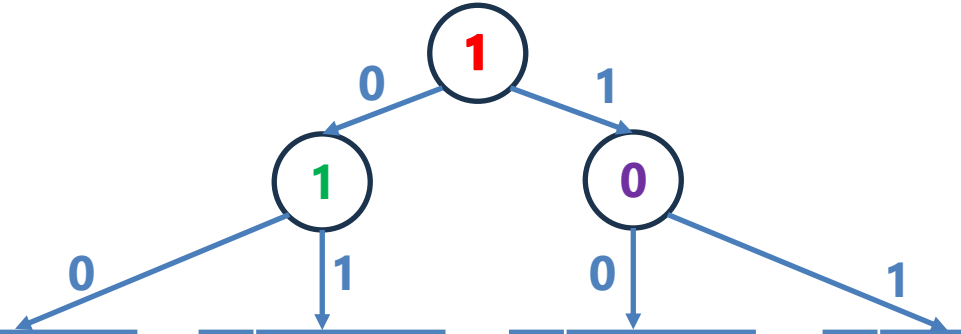


PLRU (Pseudo-LRU) Replacement

- Our read history is: 000000, 000001, 00010, 000011.

PLRU	V	Tag	V	Tag	V	Tag	V	Tag
000	0		0		0		0	

- Say our next read is to 000100.
  - Then we update PLRU bits to point to next LRU block.



PLRU	V	Tag	V	Tag	V	Tag	V	Tag
110	1	000100	1	000010	1	000001	1	000011